



NIRASHA DE MEL

Software Engineer Intern

ABOUT ME

Motivated third year IT undergraduate with a strong interest in full stack development and building end to end software solutions. Skilled in developing web applications and familiar with cloud technologies such as AWS, containerization with Docker, and basic CI/CD workflows. A fast learner and reliable team player with strong problem solving abilities. Seeking a Software Engineering Intern role to apply and grow technical skills while contributing to impactful projects.

EDUCATION

B.Sc. (Hons.) in Information Technology

University of Moratuwa

2022 - Present

- Current GPA: 3.53

G.C.E A/L: Physical Science Stream

Hillwood College Kandy

2021

- Results - BBB

LANGUAGE

- English
- Sinhala

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Kandy, Sri Lanka

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PROJECTS

AI-Powered Fitness Web App (Individual project) Ongoing

- Secure user registration, login, and profile management.
- Activity tracking with real time progress monitoring.
- AI-generated personalized fitness and health recommendations.

Technologies: Spring Boot, React, Google Gemini API, PostgreSQL, MongoDB

Salon Booking System 2024 - 2025

Second Year Software Project

Designed and implemented a robust salon booking system.

My Contribution:

- Developed client booking flow with Stripe payment integration.
- Enabled clients to book, reschedule, and cancel appointments online.
- Built salon owner dashboard to view, cancel appointments, analyze payments, and create bills

Role: Full Stack Developer

Technologies: Next.js, React, MongoDB

SafeRoute- Road Safety & Hazard Reporting Platform 2025

- Built a real-time hazard reporting platform with GPS-based reports and photo uploads.
- Implemented 25m proximity hazard detection and intelligent route planning.
- Added color-coded safety alerts and secure user authentication for community-driven road safety.

Role: Full Stack Developer

Technologies: React/Next.js, TypeScript, Tailwind CSS, Ballerina, PostgreSQL, Google Maps API, JWT Authentication

Multi-Modal AI Chatbot & Knowledge Assistant 2025

- Developed a multi-modal AI assistant with chat memory, voice I/O, and Streamlit UI.
- Implemented RAG using FAISS for web and PDF-based Q&A and summarization.
- Integrated speech-to-text and text-to-speech with LLaMA 3.2 for context-aware responses.

Technologies: Python, LLaMA 3.2, FAISS, LangChain

SOFT SKILLS

- Leadership
- Creativity
- Adaptability
- Teamwork
- Critical Thinking
- Problem solving

ACHIEVEMENTS

- AlgoXplore 1.0 – HackathonHub (2024)
- Xcelerate – SpiritX 2025 (MoraSpirit 360)
- Ballerina Coding Challenge(IEEE & WSO2, 2024)

EXTRACURRICULARS & INTERESTS

- University-level badminton player
- Organizing committee member –

Monster Mash (Rotaract, UoM) /
World Peace Day Virtual Debate
(Rotaract, UoM)

○	Automobile Service Management System	2025
	Designed and implemented a full-stack automobile service management application. My Contribution:	
	• Developed a customer dashboard using React with features for service booking, modification requests, and real-time progress tracking, integrating a generative AI-powered chatbot for customers to check available service slots.	
	Technologies: React, Node.js, Express.js, MongoDB	
○	PrimePlay - Sports Streaming App	2025
	• Built a mobile sports app with secure authentication, live match tracking, and real-time updates. • Added player profiles, favorites management, and advanced search/filtering. • Implemented dark/light mode, persistent data storage, and responsive UI with smooth animations. • Designed intuitive navigation across multiple sports categories for better user experience.	
	Technologies: React Native, Expo, TypeScript, Tailwindcss	
○	Aquatica – Automated Home Aquarium Management System	2023-2024
	First Year Hardware Project - Microcontroller Based Application Development	
	• Designed and developed an automated aquarium management system for temperature monitoring, water clarity control, and scheduled fish feeding. • Implemented automatic water refilling when clarity fell below a defined threshold. • Added day-night light cycle automation for consistent tank conditions. • Delivered real-time updates through an integrated mobile app.	
	Technologies: C++ (Arduino), IoT, Mobile App	

TECHNICAL SKILLS

○	Web Development
	HTML CSS Tailwind CSS Next.js React.js Spring Boot Three.js
	Programming Languages
	C Java TypeScript JavaScript Python
	Database
	MongoDB MySQL PostgreSQL
	Cloud
	AWS CI/CD Pipelines (GitHub Actions) Docker
	Others
	REST APIs Agile Stripe Payment Integration IoT Arduino

REFERENCES

Dr. Supunmali Ahangama Senior lecturer, University of Moratuwa Phone: +94 71 486 6652 Email : supunmali@uom.lk	Mr. Avishka Rajapaksha Software Engineer, eight25 Phone: +94 71 376 8901 Email : avishka.i@eight25media.com
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